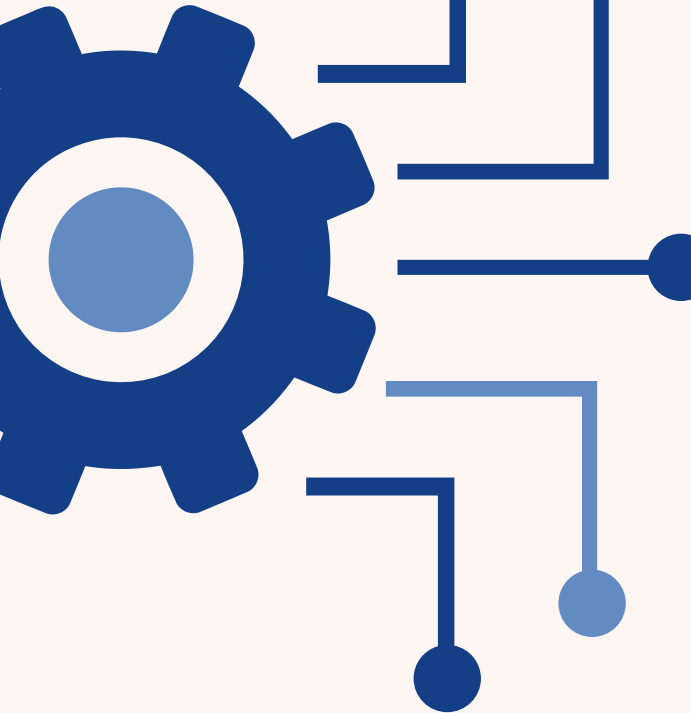
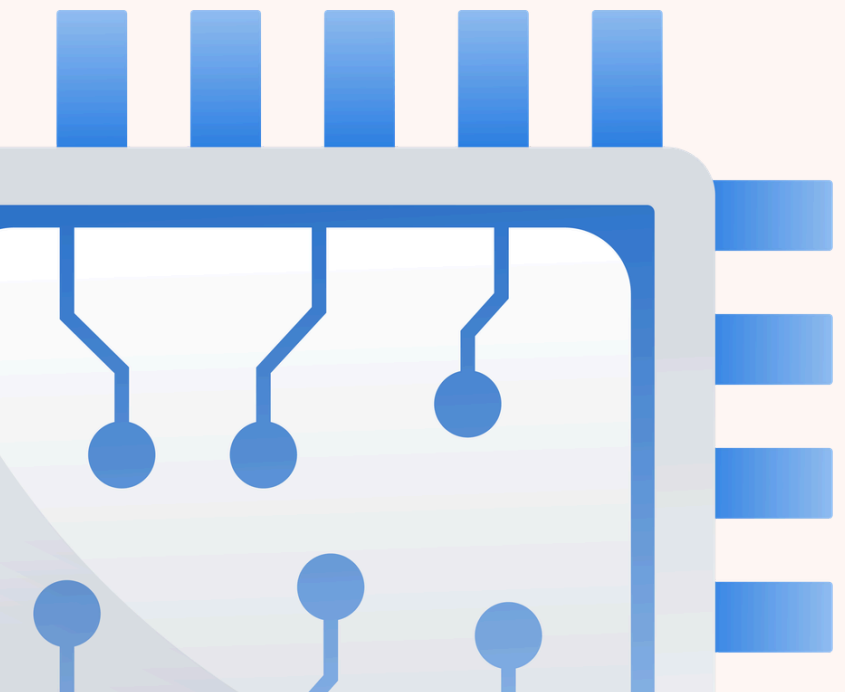




HW1: FARKLE

This is the continuation of my Farkle Project made in Java from HW 1! It contains all of the basic functionality and now some quality-of-life improvements to the game. New rerolling and improved scoring mechanics have been added, as well as the bonus “Hot Hands” occurrence.

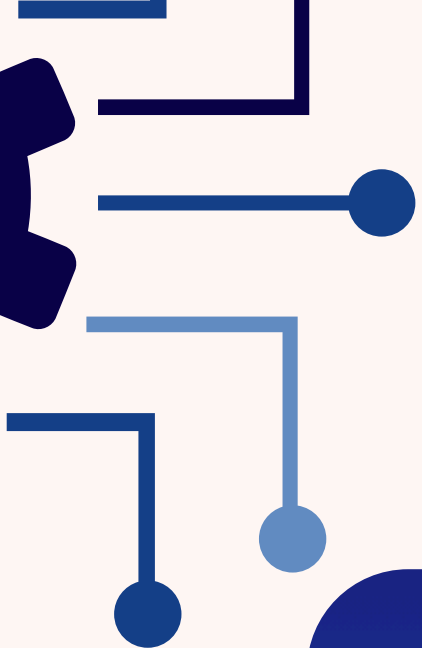


By Jett Marleau





PROCESS



Class and design Structure

This project used updated versions of the scoring and turn class structures from the last project, along with classes that allow better developer testing, and a more exciting UI with more interactivity.

Design Challenges

My biggest issue was revising my previous Scoring mechanics to be more in line with the traditional in class version of Farkle instead of the online version. This required major reworks of many systems and some complete overhauls of my previously designed functions. It also required rigorous testing of the game which was slow and tedious.

Future Changes

With more time, I would go back and revise the “Hot Hands” mechanics. As of now, it works properly, but it can be a little unclear to new users. I would go back and do more testing to make it more intuitive.



- SoftwareDev
- JettMarleauHW1
 - Summaries
 - Dice.java
 - Farkle.java
 - README.md
 - Score.java
 - SoftwareDev.code-workspace U
 - StartBanner.java
 - TestHands.java
 - Turn.java

```
rm: cannot remove '/etc/post-install/01-devices.post': Permission denied
rm: cannot remove '/etc/post-install/03-mtab.post': Permission denied
rm: cannot remove '/etc/post-install/06-windows-files.post': Permission denied
rm: cannot remove '/etc/post-install/99-post-install-cleanup.post': Permission denied
```

```
Jett Marleau@Jetts MINGW64 ~/OneDrive - Gonzaga University/Desktop/SoftwareDev (main)
```

```
$ cd JettMarleauHW1/
```

```
Jett Marleau@Jetts MINGW64 ~/OneDrive - Gonzaga University/Desktop/SoftwareDev/JettMarleauHW1 (main)
```

```
$ java Farkle.java
```

```
*****
*                               *
*           Zag Farkle by Jett Marleau           *
*                               *
*           Copyright 2026                       *
*                               *
*****
```

```
Enter a player name: Jett
Jett, it's your turn! Rolling dice...
```

```
Current Hand:[1, 5, 2, 6, 5, 3]
Occurence of each die value (1 - 6) : [1, 1, 1, 0, 2, 1]
```

```
***** Current Hand and Meld *****
```

Die	Hand	Meld
-----+-----		
(A)	1	
(B)	5	
(C)	2	
(D)	6	
(E)	5	
(F)	3	

```
(K) Keep meld score and end turn
(R) Reroll dice in hand
(Q) Quit Game
```

```
Current Score: 0
```

```
Enter the Letter for your Choice: █
```



UNIT TESTS

I made a few different unit tests that could check for simple combos like straights, 5 of a kind, triple pairs, and, of course, a Farkle. I made a testHands class that created consistent arrays with these combos that could be fed into main as a test case. I could then check to make sure the score output was correct. I could also check my “Hot Hands” mechanics consistently this way as the test array containing a straight used all six dice in the meld every time.

